QuantAlpha — Systematic Trading Tearsheet

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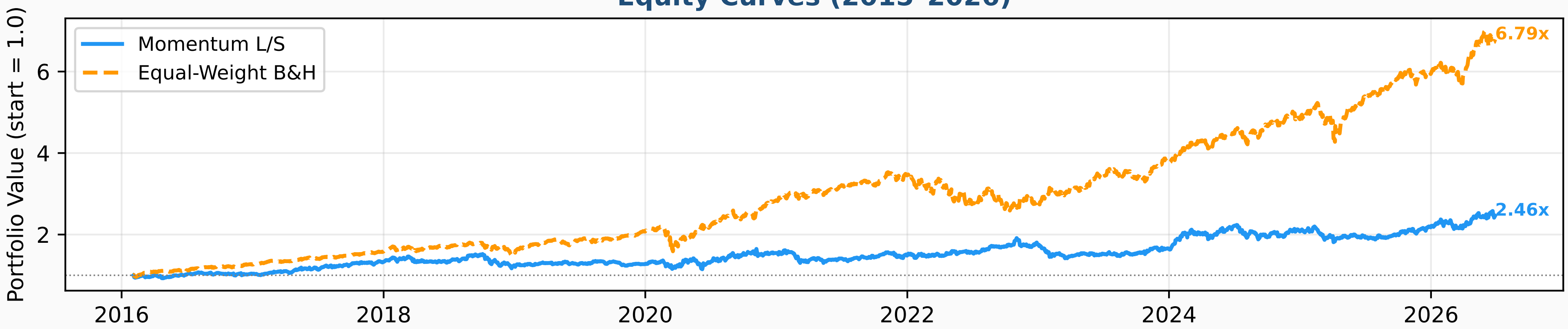
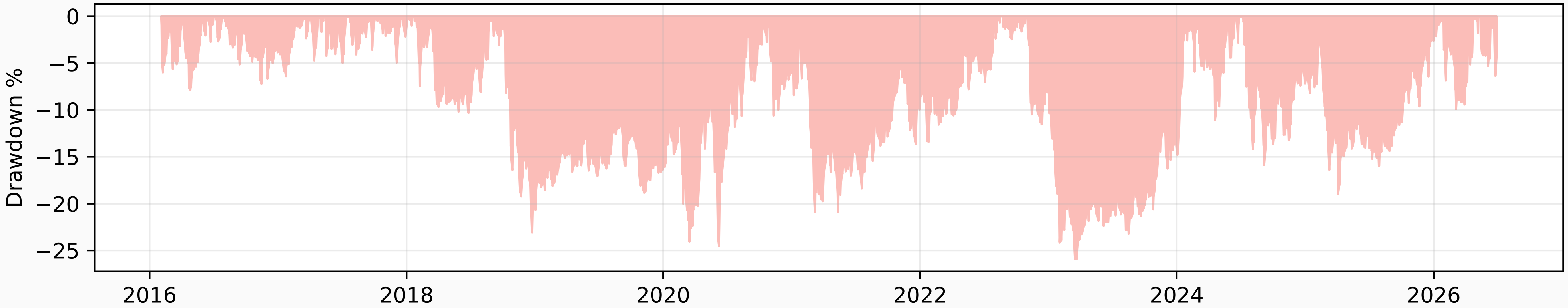
Equity Curves (2015–2026)



Strategy Drawdown

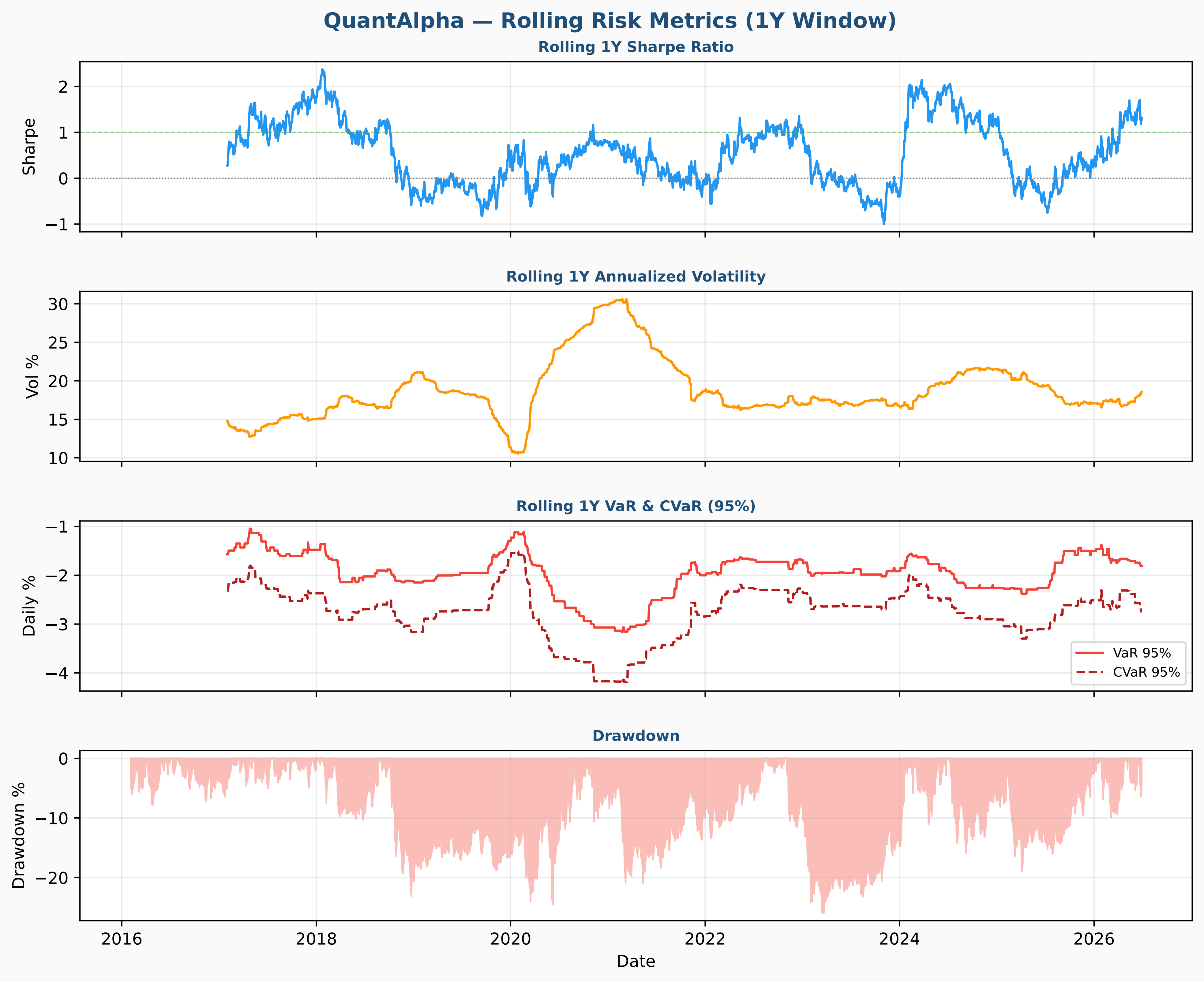


Performance Summary

	Momentum L/S	Equal-Weight
CAGR	9.08%	20.27%
Sharpe	0.55	1.19
Max DD	-25.94%	-28.20%
Ann. Vol	19.08%	16.70%
Calmar	0.35	0.72

Tail Risk (Daily)

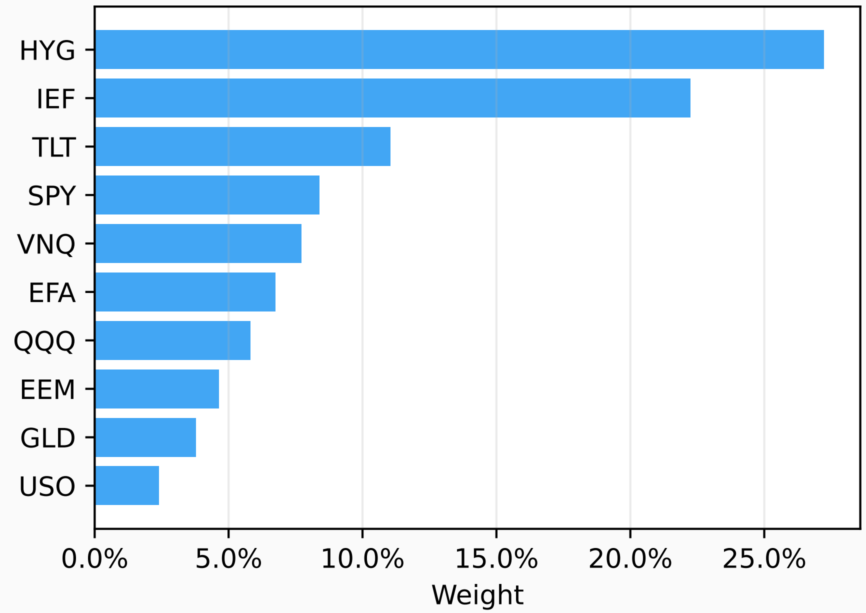
	Value
VaR 95%	-2.00%
VaR 99%	-3.43%
CVaR 95%	-2.89%
CVaR 99%	-4.31%
Skewness	-0.422
Kurtosis	2.774



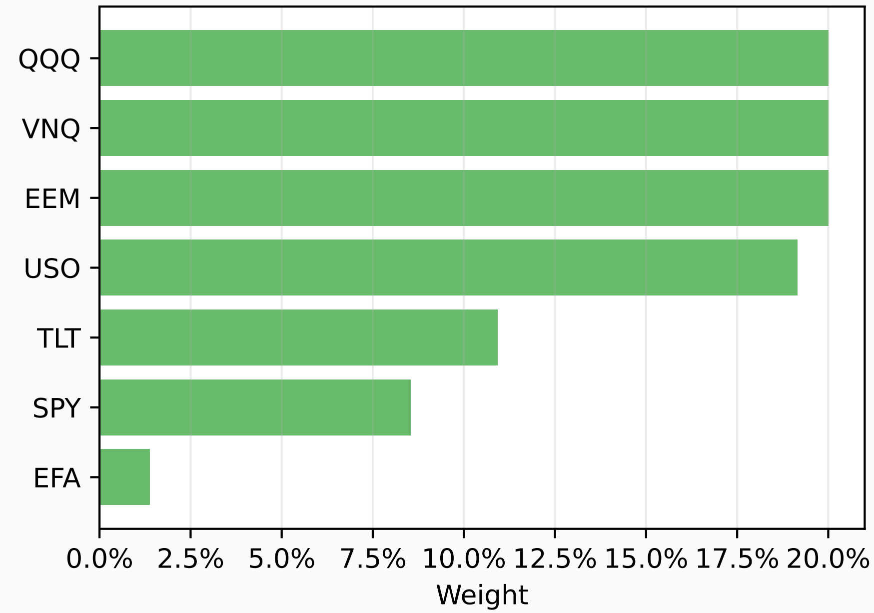
QuantAlpha — Rolling Risk Metrics (1Y Window)
Rolling 1Y Sharpe Ratio
Sharpe
2
1
0
−1
Rolling 1Y Annualized Volatility
Vol %
30
25
20
15
10
Rolling 1Y VaR & CVaR (95%)
Daily %
−1
−2
−3
−4
VaR 95%
CVaR 95%
Drawdown
Drawdown %
0
−10
−20
2016
2018
2020
2022
2024
2026
Date

QuantAlpha — Portfolio Optimization

Risk Parity Weights
(top 10, inverse vol)



Max Sharpe Weights
(mean-variance, long-only)



Weight Comparison

	Risk Parity	Max Sharpe
SPY	8.4%	8.5%
QQQ	5.8%	20.0%
GLD	3.8%	0.0%
TLT	11.0%	10.9%
EEM	4.6%	20.0%
EFA	6.7%	1.4%
HYG	27.2%	0.0%
VNQ	7.7%	20.0%
USO	2.4%	19.2%
IEF	22.2%	0.0%

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RISK PARITY
Weight inversely proportional to vol.
Lower-vol assets (bonds, gold) receive
more weight. Equal risk contribution.
Used by: Bridgewater All Weather.

MAX SHARPE (MEAN-VARIANCE)
Maximizes return per unit of risk.
Sensitive to return estimates — tends
to concentrate in recent winners.
Known limitation: estimation error.

KNOWN LIMITATION
Both optimizers use in-sample data.
Out-of-sample performance degrades
due to estimation error. Production
use requires rolling reestimation.
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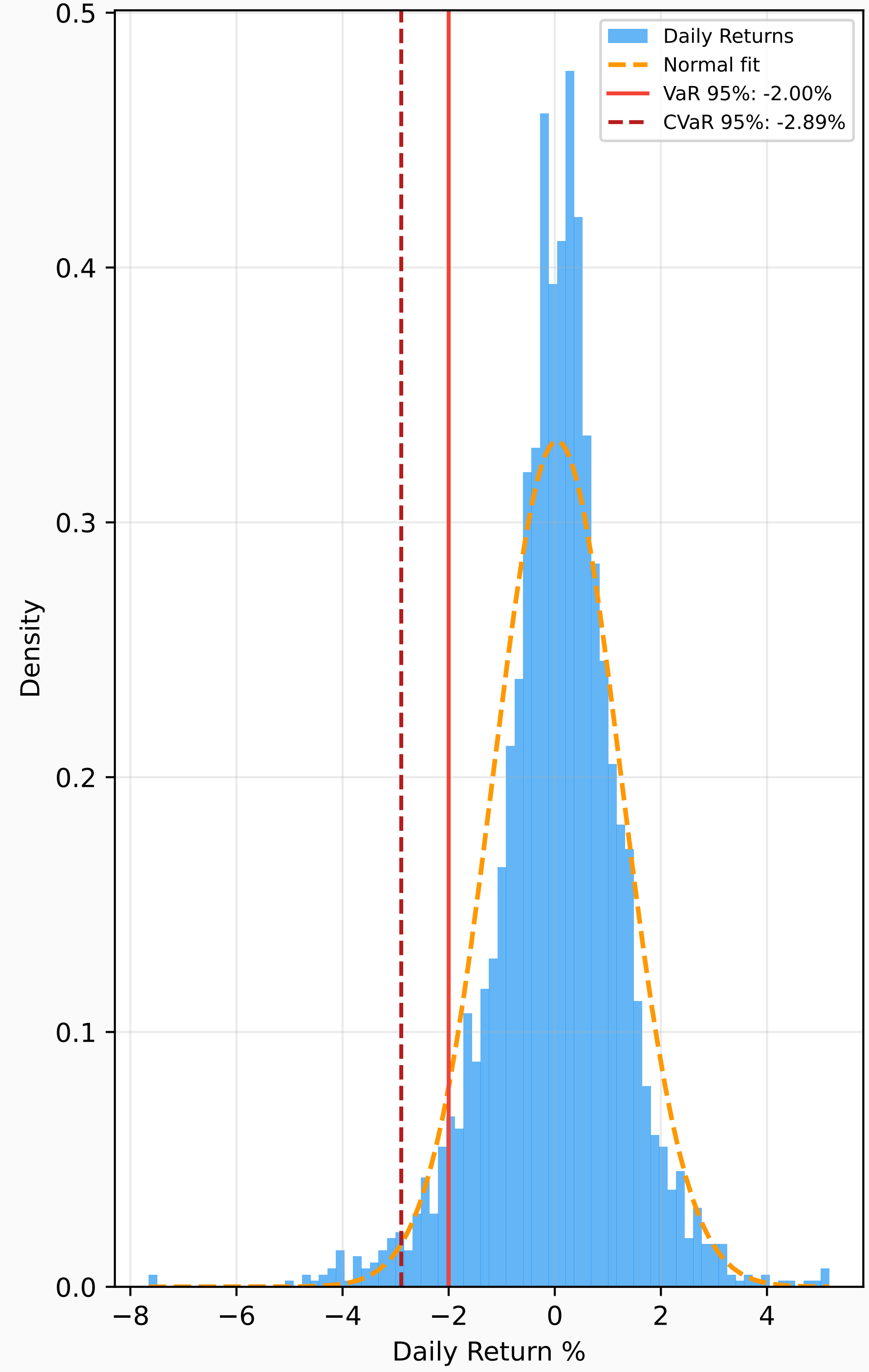
QuantAlpha — Full Risk Report

Complete Risk Metrics

	Momentum L/S
— RETURN METRICS —	
Total Return	146.41%
CAGR	9.08%
Ann. Volatility	19.08%
Sharpe Ratio	0.55
— DRAWDOWN —	
Max Drawdown	-25.94%
Calmar Ratio	0.35
— TAIL RISK (daily) —	
VaR 95% Historical	-2.00%
VaR 99% Historical	-3.43%
VaR 95% Parametric	-1.94%
VaR 95% Cornish-Fisher	-2.01%
CVaR 95% (Exp. Shortfall)	-2.89%
CVaR 99% (Exp. Shortfall)	-4.31%
— DISTRIBUTION —	
Skewness	-0.422
Excess Kurtosis	2.774
Best Day	5.17%
Worst Day	-7.65%
% Positive Days	53.6%

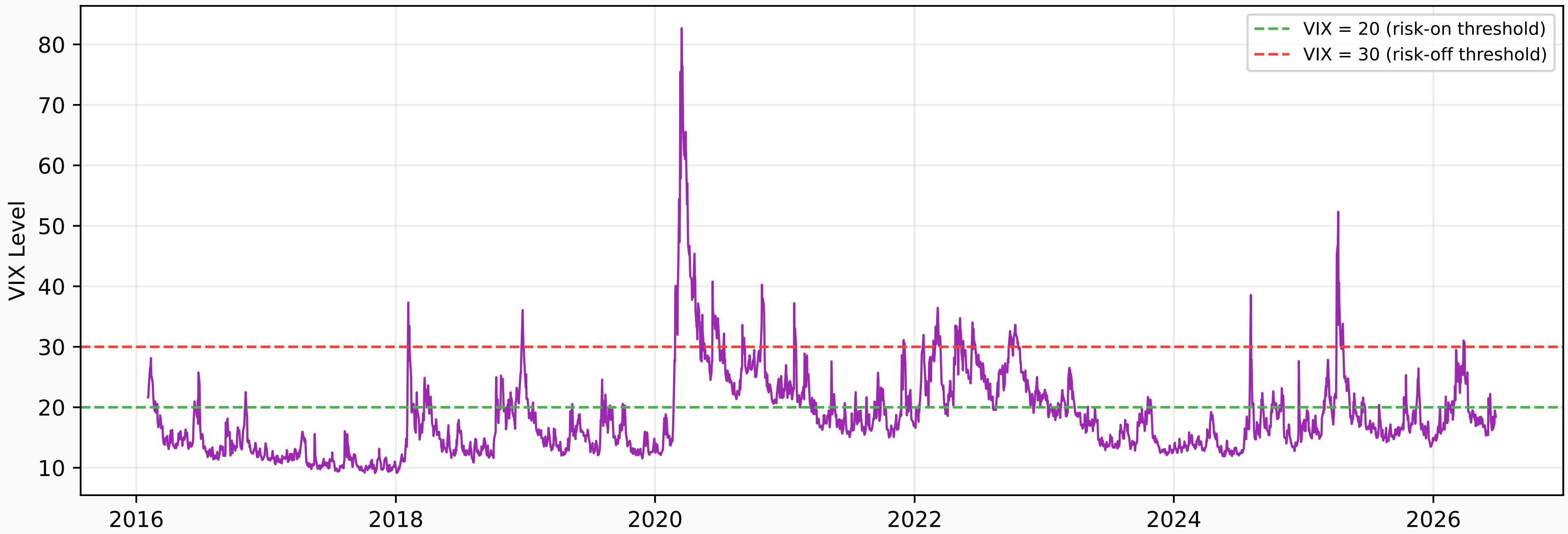
Return Distribution vs Normal



QuantAlpha — VIX Volatility Regime Analysis

How momentum performance depends on the volatility environment

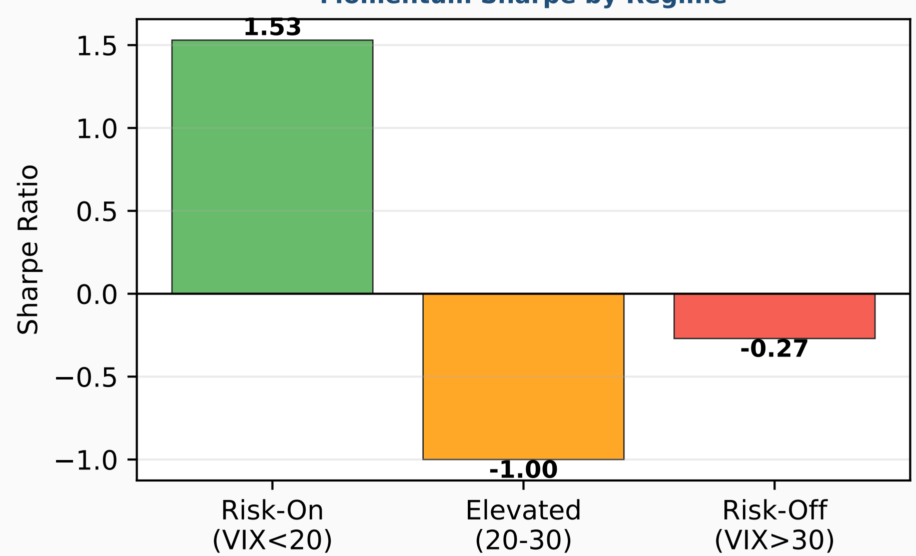
VIX Index 2015–2026 with Regime Thresholds



Performance by VIX Regime

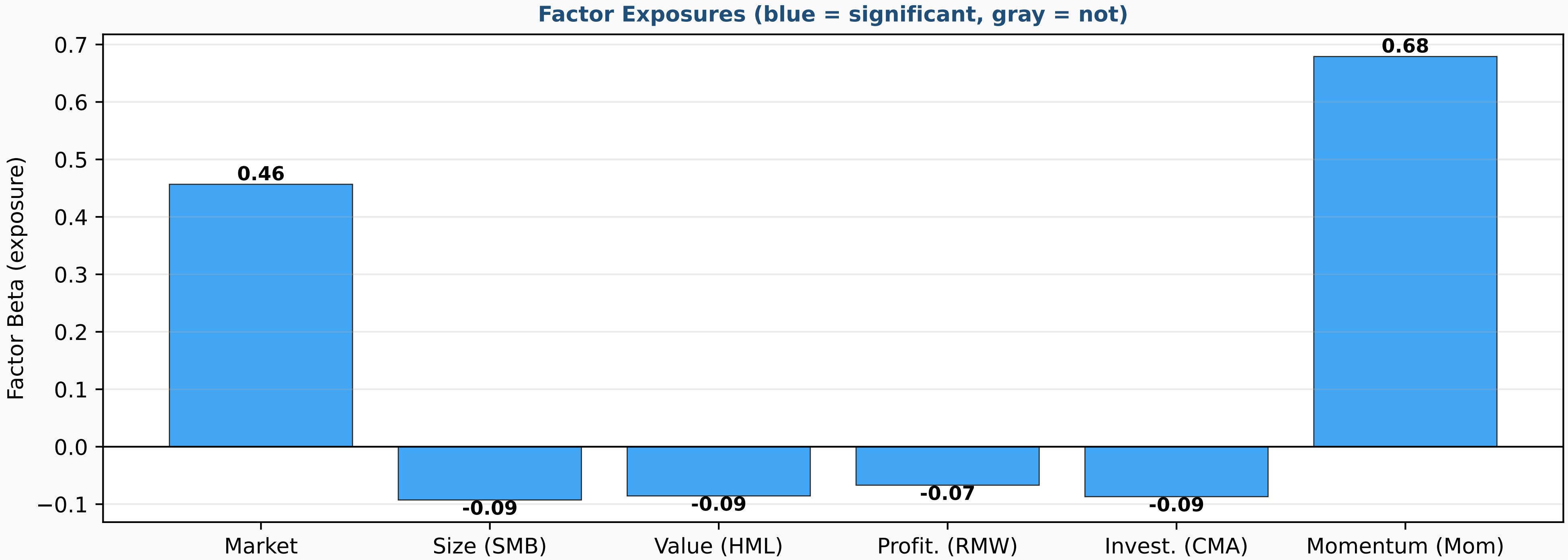
	Days	% Time	Ann. Return	Ann. Vol	Sharpe
risk_on	1813	69.4%	23.93%	15.63%	1.53
elevated	647	24.8%	-23.25%	23.17%	-1.00
risk_off	154	5.9%	-8.80%	32.02%	-0.27

Momentum Sharpe by Regime



QuantAlpha — Factor Exposure Analysis

Fama-French 5-Factor + Momentum decomposition of strategy returns



Factor Loadings

	Beta	t-stat
Market	+0.457	33.52
Size (SMB)	-0.093	-3.66
Value (HML)	-0.086	-3.77
Profit. (RMW)	-0.067	-2.16
Invest. (CMA)	-0.087	-2.31
Momentum (Mom)	+0.679	46.12

ANNUALIZED ALPHA
-0.21%
(t-stat -0.06)
NOT SIGNIFICANT

R-squared: 0.609
61% of returns explained
by the six factors

CONCLUSION
The strategy is a near-replica of the momentum factor (β=0.68) with unintended market exposure (β=0.46). No true alpha.